*Tick Tock Bomb *

.....__FUN WEEKENED ACTIVITY__.....

##OVERVIEW OF THE ACTIVITY :-

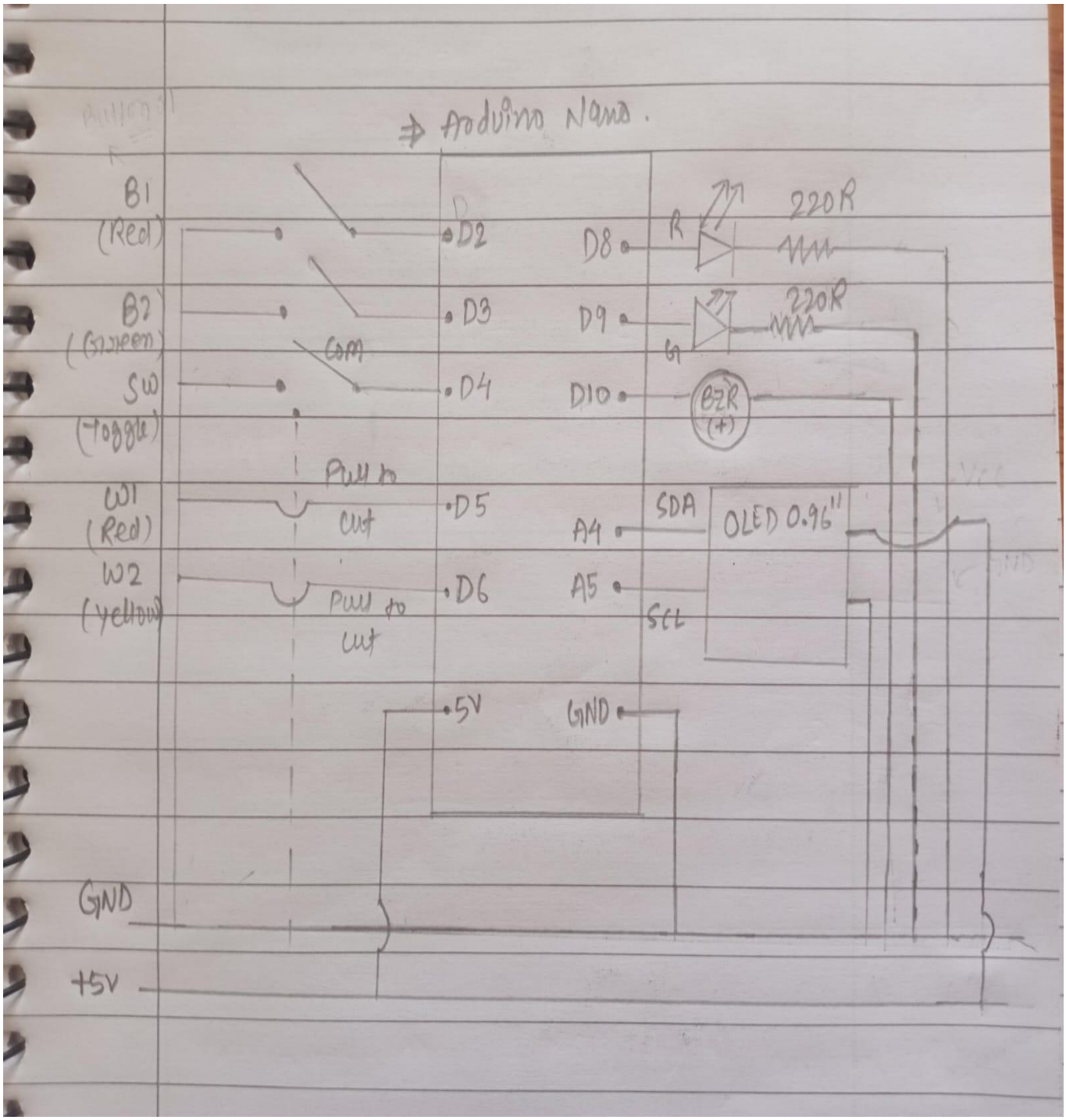
Imagine a movie scene where the hero has to defuse a bomb before it explodes. That's exactly what we're building — but as a fun game!

The Tick Tock Bomb is a timed puzzle built around an Arduino Nano, where the player has to complete a series of challenges before a 60-second countdown hits zero. When you power it on, a tiny 0.96-inch OLED screen greets you with "BOMB DEFUSAL — Press RED to start." Once you hit the red button, the countdown 3-2-1 start and then it display five randomly shuffled challenges at you, one at a time. These could be anything from pressing the red or blue button, pressing blue three times in a row, flipping a toggle switch, or physically pulling out the correct jumper wire from the breadboard. Wrong actions subtract 10 seconds as a penalty. Cutting the decoy wire triggers instant failure. The buzzer beeps faster as time runs out, creating genuine tension and excitement.

##Why this project?

It teaches digital I/O, debouncing, edge detection, I2C communication, PWM audio, game state machines, and timing — all fundamental embedded systems concepts — while being genuinely fun to play and demonstrate.

##CIRCUIT DIAGRAM :



##Pin Connection Table :

Arduino Pin	Component
5V	OLED VCC, Breadboard + rail
GND	OLED GND, Breadboard − rail
D2	Button 1 (Red)
D3	Button 2 (Blue)
D4	Toggle Switch
D5	Red Jumper Wire
D6	Yellow Jumper Wire
D8	Red LED (anode)
D9	Green LED (anode)
D10	Passive Buzzer (+)
A4 (SDA)	OLED SDA
A5 (SCL)	OLED SCL

##Bill of Materials (BOM) :

Sr.no.	Component's Name	Quantity	Price (in Rs.)
01	Arduino Nano (CH340 clone)	1	₹299
02	Mini Breadboard (400 tie-points)	1	₹49
03	0.96" OLED Display (SSD1306 I2C)	1	₹189
04	Tactile Push Button (12mm)	2	₹5 each (₹10)
05	SPDT Toggle Switch	1	₹15
06	Male-to-Male Jumper Wires (set of 20)	1	₹49
07	Red LED (5mm)	1	₹3
08	Green LED (5mm)	1	₹3
09	Passive Buzzer	1	₹19
10	220Ω Resistor (1/4W)	2	₹2 each (₹4)
11	USB Mini-B Cable	1	₹29
		TOTAL(approx.)	₹669

##How It Works - Complete Explanation :

The game operates as a finite state machine with five states that transition based on player actions and timer events:

1. **WELCOME —** The OLED shows "BOMB DEFUSAL" with a bomb icon and "Press RED to start". The system waits for Button 1 (red) to be pressed. All LEDs are off, buzzer is silent.
2. **COUNTDOWN —** A dramatic 3-2-1-GO! countdown appears on the OLED with a beep on each number. During this time, the game generates a randomized sequence of 5 defusal steps using a Fisher-Yates shuffle algorithm.
3. **PLAYING —** The main game loop. The OLED shows: a shrinking timer bar at the top, the current step number (e.g., "3/5"), the current clue in large text (e.g., "CUT THE RED WIRE"), and the score. The buzzer beeps at intervals that decrease as time runs out. The red LED blinks faster in the final seconds.
4. **WIN —** If the player completes all 5 steps before time runs out, the green LED lights up, a victory melody plays through the buzzer, and the OLED shows "DEFUSED!" with the final score (including a time bonus). Press red button to restart.
5. **LOSE —** If the timer reaches zero or the player cuts the decoy (yellow) wire, the red LED goes solid, an explosion sound effect plays, and the OLED shows "BOOM!" with explosion rays. Press red button to restart.

Software Setup Guide:

1 Install Arduino IDE

Download Arduino IDE 2.x from arduino.cc/en/software. Install it with default settings. Windows, macOS, and Linux are all supported.

2 Install the U8g2 Library

- Open Arduino IDE and go to Sketch → Include Library → Manage Libraries
- Search for "U8g2" and install "U8g2 by oliver" (any recent version works)
- Close and reopen Arduino IDE after installation

Important: Make sure your Sketchbook location (File → Preferences) points to a simple path like C:\Arduino — paths with special characters, OneDrive, or non-English folder names can prevent libraries from being found.

3 Board Settings

- Board: Arduino Nano
- Processor: ATmega328P (Old Bootloader) — use this for clone boards
- Port: Select the COM port that appears when you plug in the Nano via USB

4 Upload the Code

- ✓ Open the code file in Arduino IDE
- ✓ Click the checkmark button (Verify) to compile — it should show "Done compiling" with no errors
- ✓ Click the arrow button (Upload) to flash the code onto the Nano
- ✓ The OLED should immediately show the welcome screen once upload completes

#ARDUINO CODE: https://github.com/Nitinraj20/Tick-Tock-Bomb.git